

PhoenixHRMS

SUMMARY

The Integrations Platform provides a scalable, secure, and enterprise-grade connectivity layer that enables PhoenixHRMS to operate as part of a broader enterprise ecosystem.

DOCUMENT METADATA

Version: 1.0 Status: Draft Module ID: INT Document Type: Platform Specification Related Documents: 13-api-standards.md; 14-openapi-spec.md; 06-system-architecture.md; 26-ai-copilot-module.md

PLATFORM OVERVIEW

Purpose: The Integrations Platform provides a unified framework for connecting PhoenixHRMS with internal systems, third-party applications, cloud platforms, government services, and enterprise ecosystems. Enables: API Integrations; Event Integrations; Webhook Integrations; Data Synchronization; Workflow Automation; Marketplace Extensions
Business Objectives: Enable ecosystem connectivity; Reduce implementation effort; Accelerate enterprise adoption; Improve interoperability; Support partner integrations; Enable automation

INTEGRATION VISION

PhoenixHRMS -> Integration Platform -> Enterprise Ecosystem -> Unified Digital Workplace

INTEGRATION ARCHITECTURE

Applications -> API Gateway -> Integration Layer -> Event Bus -> Adapters -> External Systems Design Principles: API First; Event Driven; Cloud Native; Secure By Default; Extensible; Multi-Tenant Aware

INTEGRATION CATEGORIES

HR Systems: Payroll Systems; HRIS; Benefits Platforms; Talent Platforms ERP Systems: SAP; Oracle ERP; Microsoft Dynamics; NetSuite Finance Systems: QuickBooks; Xero; Tally; Zoho Books Identity Providers: Microsoft Entra ID; Okta; Google Workspace; OneLogin; Ping Identity Collaboration Platforms: Microsoft Teams; Slack; Google Chat; Zoom Government Platforms: Tax Portals; PF Portals; ESI Portals; Labor Portals; Compliance Platforms

API PLATFORM

Purpose: Primary integration mechanism. API Types: REST APIs; Webhook APIs; Internal APIs; Partner APIs; Public APIs Standards: OpenAPI 3.1; OAuth 2.1; JWT; JSON; HTTPS

API GATEWAY

Responsibilities: Authentication; Authorization; Rate Limiting; Logging; Caching; Request Routing; Monitoring
Features: API Keys; OAuth Tokens; Request Validation; Traffic Management

EVENT-DRIVEN ARCHITECTURE

Purpose: Real-time integration. Flow: Business Event -> Event Bus -> Subscribers -> Actions Benefits: Loose Coupling; Scalability; Real-Time Processing; Resilience

EVENT BUS

Supported Events: Employee Created; Employee Updated; Leave Approved; Payroll Completed; Candidate Hired; Course Completed; Asset Assigned; Document Uploaded Event Format: { "event_type": "", "event_id": "", "timestamp": "", "tenant_id": "", "payload": {} }

WEBHOOK PLATFORM

Purpose: Push data to external systems. Webhook Events: Employee Created; Attendance Updated; Leave Approved; Payroll Processed; Recruitment Status Changed; Performance Review Completed Delivery: At Least Once Features: Retries; Signing; Validation; Monitoring

WEBHOOK SECURITY

Controls: Webhook Secrets; Request Signing; Replay Protection; IP Whitelisting; TLS Enforcement

DATA SYNCHRONIZATION

Synchronization Types: Real-Time; Scheduled; Batch; Manual Modes: One-Way; Two-Way; Master-Slave; Bidirectional

INTEGRATION ADAPTERS

Purpose: Normalize external systems. Adapter Types: HR Adapter; ERP Adapter; Finance Adapter; Identity Adapter; Communication Adapter; Custom Adapter

IDENTITY INTEGRATIONS

Single Sign-On: SAML 2.0; OAuth 2.1; OpenID Connect Providers: Microsoft Entra ID; Okta; Google Workspace; Auth0 Features: Just-In-Time Provisioning; SCIM; Role Mapping; User Sync

SCIM PROVISIONING

Supported Operations: Create User; Update User; Deactivate User; Group Sync; Role Sync Standards: SCIM 2.0

PAYROLL INTEGRATIONS

External Payroll Systems: ADP; Paychex; SAP Payroll; Oracle Payroll; Regional Payroll Systems Data Flows: Employee Data; Attendance Data; Leave Data; Payroll Results

ERP INTEGRATIONS

Supported Systems: SAP; Oracle ERP; Microsoft Dynamics; NetSuite; Infor Data Exchange: Employees; Cost Centers; Departments; Projects; Payroll Costs

ACCOUNTING INTEGRATIONS

Supported Systems: QuickBooks; Xero; Tally; Zoho Books Integration Areas: Payroll Journals; Expense Data; Reimbursements; Asset Costs

BANKING INTEGRATIONS

Use Cases: Salary Payments; Vendor Payments; Loan Processing; Reimbursement Processing Features: Payment File Generation; Payment Status Tracking; Bank Confirmation

RECRUITMENT INTEGRATIONS

Job Boards: LinkedIn; Indeed; Naukri; Monster; Glassdoor Features: Job Publishing; Candidate Import; Application Sync

LEARNING INTEGRATIONS

Standards: SCORM; xAPI; AICC; LTI Content Providers: LinkedIn Learning; Udemy Business; Coursera; Skillsoft

COMMUNICATION INTEGRATIONS

Email: SMTP; Microsoft 365; Google Workspace; SendGrid; SES Messaging: Twilio; WhatsApp Business; Vonage; MessageBird

DOCUMENT INTEGRATIONS

Storage Providers: AWS S3; Azure Blob Storage; Google Cloud Storage Signature Providers: DocuSign; Adobe Sign; Dropbox Sign

ANALYTICS INTEGRATIONS

BI Platforms: Power BI; Tableau; Looker; Qlik; Metabase Methods: API; Database Views; Data Exports; Data Warehouse Connectors

AI INTEGRATIONS

LLM Providers: OpenAI; Anthropic; Google; Azure OpenAI AI Services: OCR; Translation; Speech To Text; Document Analysis

INTEGRATION MARKETPLACE

Purpose: Extension ecosystem. Components: Connectors; Extensions; Plugins; Partner Apps; Certified Integrations

MARKETPLACE GOVERNANCE

Certification Process: Security Review; Performance Review; Compliance Review; Documentation Review Partner Levels: Verified; Certified; Premium

INTEGRATION MONITORING

Metrics: Requests; Errors; Latency; Retries; Failures; Success Rate Dashboards: Real-Time Monitoring; Historical Trends; Integration Health

ERROR HANDLING

Strategies: Retry; Dead Letter Queue; Alerting; Fallback; Compensation Actions Error Categories: Validation; Authentication; Network; Business Rules; System Errors

INTEGRATION APIS

Connectors: GET /integrations; POST /integrations Webhooks: GET /webhooks; POST /webhooks Events: GET /events; POST /events Marketplace: GET /marketplace/apps

DATABASE TABLES

Integration Domain: integrations; integration_configs; integration_credentials Webhook Domain: webhooks; webhook_deliveries; webhook_logs Event Domain: events; event_subscriptions; event_deliveries Marketplace Domain: marketplace_apps; marketplace_installations Monitoring Domain: integration_metrics; integration_alerts

PERMISSIONS MATRIX

integration.view; integration.manage; integration.install; integration.configure; integration.monitor; webhook.manage; event.manage; marketplace.manage

SECURITY REQUIREMENTS

Encryption At Rest; Encryption In Transit; Credential Vaulting; Secret Rotation; RBAC Enforcement; Tenant Isolation; Audit Logging

AUDIT REQUIREMENTS

Integration Created; Integration Updated; Webhook Created; Webhook Delivered; Connector Installed; Marketplace App Installed; Credentials Rotated Retention: 7 Years Minimum

COMPLIANCE REQUIREMENTS

GDPR; SOC 2; ISO 27001; Data Residency; Cross-Border Data Controls; Vendor Risk Management

DASHBOARD WIDGETS

Active Integrations; Webhook Deliveries; Failed Integrations; Event Throughput; Marketplace Apps Installed; Integration Health Score

PERFORMANCE REQUIREMENTS

API Response: < 500ms Webhook Delivery: < 5 Seconds Event Processing: 100,000 Events; < 60 Seconds Integration Availability: 99.99%

FUTURE ENHANCEMENTS

Version 2: iPaaS Platform; Low-Code Integration Builder; Workflow Designer; Event Marketplace; Enterprise Service Bus; Integration AI Assistant; Autonomous Integrations; Partner Developer Portal

SUCCESS METRICS

Integration Success Rate: > 99.9% Webhook Delivery Success: > 99.5% Marketplace Adoption: > 50% Integration Setup Time: < 30 Minutes

CONCLUSION

The platform integrates APIs, events, webhooks, identity systems, ERP systems, finance systems, communication providers, AI services, and marketplace extensions into a unified integration framework while maintaining reliability, performance, security, compliance, and tenant isolation.